

Yash Prakash Mishra

Senior Backend Engineer — Node.js · TypeScript · PostgreSQL · Real-Time Systems

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Open to relocation to Berlin — hybrid · English: professional fluency

PROFESSIONAL SUMMARY

Senior backend engineer with **5+ years in Node.js and TypeScript on PostgreSQL**, and a track record of **deleting administrative work**: 80% less manual error resolution in payments, 40% less manual intervention in loan servicing, a fully manual loan-closure process replaced outright for **10,000+ active loans**, and 10+ hours a week returned to an operations team. Built a **provider-agnostic integration layer** across four payment providers with runtime selection, **circuit-breaker fallback**, idempotent retries and event deduplication — designed for the reality that third-party services fail mid-request. Comfortable operating what I ship: Kafka event pipelines, **real-time low-latency services** on Socket.io and Redis, ELK and distributed tracing, Docker, GitHub Actions, and zero-downtime PostgreSQL and ClickHouse migrations. I own problems from design through production support.

CORE SKILLS

Languages & Runtime	TypeScript, JavaScript (ES6+), Node.js, SQL, C++
Backend & APIs	Express.js, NestJS, Fastify, RESTful API design, structured request/response contracts, input validation & schema guards, OpenAPI/Swagger, JWT auth, idempotent API design, webhooks
PostgreSQL & Data	PostgreSQL (data modeling, zero-downtime migrations, query optimization), ClickHouse, MySQL, MongoDB, Redis, Sequelize, Prisma
Real-time	Socket.io, Server-Sent Events, Redis-backed live state, low-latency concurrent sessions, graceful degradation under load
Reliability & Fallbacks	Circuit breakers (opossum), retry with backoff (axios-retry), provider failover, event deduplication, idempotency, self-healing pipelines, reconciliation
Event-Driven	Apache Kafka, BullMQ, AWS SQS, event-driven architecture, webhook processing at scale
Cloud & CI/CD	AWS (EC2, S3, CloudFront, SQS, IAM, KMS), Docker, Kubernetes (working knowledge), GitHub Actions, Jenkins, Apache Airflow
Observability & Debugging	ELK Stack, structured logging, distributed tracing, debugging microservices in production, on-call and incident reduction
Testing	Jest, Supertest, Mocha, unit & integration testing, Postman
Architecture	Microservices, distributed systems, multi-tenant architecture, saga pattern, Strategy / Factory / Singleton / Observer, HLD & LLD
Frontend (secondary)	React.js, Next.js, Redux

PROFESSIONAL EXPERIENCE

Novatr — Senior Software Engineer (Backend Developer → Senior Software Engineer) **Aug 2023 – Present**

Remote · Y Combinator-backed SaaS platform, 24/7 on AWS · Led a team of 5 engineers

Stack: Node.js, TypeScript, NestJS, Express.js, PostgreSQL, ClickHouse, Redis, Apache Kafka, BullMQ, Socket.io, AWS (EC2, S3, SQS, KMS, IAM), Docker

Third-party integrations, fallbacks & failure modes

- Architected a **provider-agnostic integration layer** across four payment providers (Stripe, Razorpay, Affirm, Jodo) using **Strategy and Factory patterns for runtime provider selection**, so a new provider onboards without touching core logic; owned each integration end-to-end from POC to production.
- Added **circuit breakers** (opossum) and **retry with backoff** (axios-retry) for automatic failover when a provider degrades or times out — the checkout path stays available through upstream outages instead of failing with it.
- Designed a centralised webhook handler for all four providers with self-healing retries, **event deduplication** and Kafka-based status reconciliation, **cutting manual error resolution by 80%**.
- Made every payment path **idempotent**, so retries, duplicate provider callbacks and client re-submissions are safe under concurrent load.

Real-time & event-driven systems

- Built a **real-time quiz engine** (Node.js, **Socket.io**, Redis) for live cohort sessions, tuned for **low-latency concurrent interactions** with graceful degradation when connections drop under load.
- Introduced **Kafka event pipelines** (with BullMQ for in-process jobs) for payments, user lifecycle events and transactional email, decoupling services and improving fault tolerance platform-wide.

PostgreSQL, data modeling & migrations — Novatr

- Owned a **zero-downtime migration** of payments history off a legacy system onto a ClickHouse-backed platform via Node.js ETL, with schema validation, integrity checks and a tested rollback path.
- Designed the **multi-tenant PostgreSQL data model** for the B2C-to-B2B transition — tenant-level data isolation, per-tenant configuration, enterprise onboarding workflows — and **onboarded 5+ enterprise clients** as technical point of contact.
- Built a CMS backend (NestJS, PostgreSQL) with content lifecycle workflows and versioning: **85% faster cohort creation**, and recurring on-call incidents eliminated.

Access control, review & team

- Architected **centralised RBAC across 5+ internal tools** with role inheritance, permission scoping and audit logging — **60% fewer access incidents**; plus a multi-layer **JWT** service (Passport.js) for token issuance, rotation and validation across three surfaces.
- Led 5 engineers across architecture decisions, **pull-request review** and delivery timelines; partnered with product, design and QA from discovery through production.

OLX Autos — Software Engineer

May 2022 – May 2023

Remote · Used-car financing infrastructure for Latin America — regulated, high-stakes workflows

Stack: Node.js, TypeScript, Express.js, PostgreSQL, MySQL, Sequelize, REST APIs, Docker, AWS

- Built and owned foreclosure and manual repayment flows in the Loan Management System, automating loan closure and repayment tracking for **10,000+ active loans in Chile** and **replacing a fully manual process** that the operations team had been running by hand.
- Designed **loan lifecycle state machine APIs** with **validation guards at every transition** — explicit legal states, explicit rejections, an audit trail per change — **cutting manual intervention by 40%**.
- Designed risk analysis and credit scoring workflows in the Loan Origination System, **reducing credit approval time by 30%** while improving decision auditability.
- Built a reusable PDF generation library (PDFKit, Handlebars) for programmatic invoices in loan disbursement, **returning 10+ hours a week** to the operations team; technical point of contact for the Chile team across a 9+ hour time-zone gap.

Lenskart — Software Engineer (Frontend)

Apr 2021 – Apr 2022

India's largest eyewear retailer · Exit: project concluded on successful UAE market launch

Stack: React.js, Redux, TypeScript, Next.js, Node.js (BFF), Webpack, REST APIs

- Led the rebuild of core commerce flows (homepage, cart, checkout, order tracking) in React.js, Redux and TypeScript for the UAE launch — **30% faster launch** with improved Lighthouse scores, as primary frontend point of contact for go-live.
- Defined response contracts against a **Node.js BFF** layer with backend teams, and cut initial JS payload through code splitting and bundle analysis — the work that moved me into backend engineering.

SELECTED PROJECT

BullMQ Dashboard — open-source operational tooling for distributed job pipelines

2025

Next.js (App Router), TypeScript, Node.js, Redis, ioredis, BullMQ, Express.js, Docker, SVG, Server-Sent Events · github.com/xcalibre1/bullmq-dashboard

- Problem:** Bull Board-style tools show queue state but give no operational control when a pipeline misbehaves in production — no retrigger with a corrected payload, no bulk retry, no live view.
- Built a **Next.js + TypeScript operations dashboard** with **Server-Sent Events** for live queue stats, an editable JSON payload editor, a 24-hour throughput chart in pure SVG, plus payload field search, bulk retry, promote, pause and resume — the tooling I wanted while debugging distributed job failures on call.
- API routes backed by a **Redis singleton** (ioredis) and a cached BullMQ Queue instance for near-zero per-request overhead; Docker Compose for one-command self-hosting. **Adopted by engineering teams globally.**

EDUCATION

B.Tech, Computer Science — Rajkiya Engineering College, Sonbhadra

Jul 2016 – Jun 2020

ADDITIONAL

- Open source:** author and maintainer of BullMQ Dashboard (Node.js, TypeScript), used by engineering teams globally for queue operations.
- Relocation & languages:** open to relocating to Berlin for a hybrid role; professional fluency in English; 5 years working across distributed teams in India, Latin America and the Middle East.
- Currently learning:** LLM application engineering — prompting, tool calling, structured outputs and evaluation harnesses — alongside streaming audio and real-time voice pipelines.